
Topic: Review of Probability

Important note for submissions:

- Only handwritten answers are allowed.
- The hard-copy of solutions should be submitted to me in Wednesday lecture.
- No email submissions will be entertained.
- **Submit solutions to ONLY Problems 1, 2, 5, 7, and 9.**

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1. **The Birthday Paradox.** Suppose there are 23 people in a room. Assume that each person's birthday is equally likely to fall on any of the 365 days of the year, independently of everyone else, and ignore leap years.
   (a) Compute the probability that all 23 people have distinct birthdays.
   (b) Hence compute the probability that at least two people share the same birthday.
   (c) Evaluate the probability numerically (to four decimal places) and comment on whether the result is surprising.

2. **Interpreting a Positive Test.** A disease affects 1% of the population. A diagnostic test has the following characteristics:
   - Sensitivity: 95% (the test correctly identifies an infected person with probability 0.95).
   - Specificity: 98% (the test correctly identifies a healthy person with probability 0.98).

   A randomly selected individual tests positive.
   (a) What is the probability that the individual actually has the disease?
   (b) What is the probability that the positive result is a false positive?
   (c) Briefly explain why the answer may differ from one's intuition.

3. **The Monty Hall Problem.** A game show presents three closed doors. Behind one door is a car, and behind each of the other two doors is a goat. You choose one of the three doors.

   The host, who knows where the car is located, then opens one of the two remaining doors, always revealing a goat. The host then offers you the opportunity to switch to the remaining unopened door.
   (a) What is the probability of winning the car if you keep your original choice?
   (b) What is the probability of winning the car if you always switch?
   (c) Explain your reasoning carefully.

4. **The Maximum of Two Dice.** Two fair six-sided dice are rolled independently. Let

$$M = \max(X, Y),$$

where X and Y denote the outcomes of the two dice. Determine the probability mass function of M. Verify that it is a valid probability distribution, and compute the expected value $E(M)$.

5. **Counting Sixes.** A fair six-sided die is rolled 120 times. Let X denote the number of times a six is observed. Without directly invoking the Binomial distribution, use indicator random variables to derive the following quantities:

   (a) the expected value of X,
   (b) the variance of X,
   (c) the standard deviation of X.

   Interpret each of these quantities in the context of this experiment.

6. **Bounding Probabilities.** Suppose a random variable X satisfies

$$E(X) = 50, \qquad \mathrm{Var}(X) = 64.$$

   Using Chebyshev's inequality, obtain upper bounds for the following probabilities:

   (a) $P(|X - 50| \geq 8)$,
   (b) $P(|X - 50| \geq 16)$,
   (c) $P(34 \leq X \leq 66)$.

   Comment on how the bound changes as the interval around the mean becomes wider.

7. **Properties of the Sample Mean.** Let $X_1, X_2, \ldots, X_n$ be independent and identically distributed random variables with

$$E(X_i) = \mu, \qquad \mathrm{Var}(X_i) = \sigma^2.$$

   Let

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

   denote the sample mean.

   (a) Show that $E(\bar{X}) = \mu$.
   (b) Show that $\mathrm{Var}(\bar{X}) = \sigma^2/n$.
   (c) Explain why the standard deviation of $\bar{X}$ decreases as the sample size increases.
   (d) If $\sigma = 12$, compute the standard deviation of $\bar{X}$ for $n = 9$, 36, and 144.

8. **Normal Distribution.** The heights of adult women in a certain population are approximately normally distributed with mean 162 cm and standard deviation 6 cm.

   (a) What proportion of women are taller than 170 cm?
   (b) What proportion have heights between 156 cm and 168 cm?
   (c) Determine the height that corresponds to the 90th percentile of the population.

Clearly show the standardization step in each part.

9. **Normal Distribution.** Let $X \sim N(\mu, \sigma^2)$.

   (a) Show that
   $$P(\mu - \sigma \leq X \leq \mu + \sigma) = P(-1 \leq Z \leq 1),$$
   where $Z \sim N(0, 1)$.

   (b) Hence compute the probability that a normally distributed observation lies within one, two, and three standard deviations of its mean.

   (c) Explain why these probabilities are independent of the values of μ and σ.

10. **Symmetry of normal distribution.** Suppose $X \sim N(\mu, \sigma^2)$. Without using numerical tables, express each of the following probabilities in terms of the standard normal distribution function $\Phi(\cdot)$.

    (a) $P(X > \mu)$

    (b) $P(X < \mu + 2\sigma)$

    (c) $P(\mu - \sigma < X < \mu + 2\sigma)$

    (d) Show that
    $$P(X > \mu + a) = P(X < \mu - a)$$
    for every $a > 0$.